

How Codex was used to build Sky Fundi

AI-assisted engineering under human governance — the setup, the workflow, and what it delivered.

Sky Fundi was built with an AI engineering agent ("Codex") working as a **governed team member** — not as an autocomplete. The repository itself defines how the agent must behave, every change went through the same gates a human contribution would, and a human reviewed and merged every pull request.

1 · Governance before generation

The repository's `AGENTS.md` is the agent's operating manual, versioned like code. It binds the agent to:

- **Scope control** — the smallest cohesive change; no speculative features, no drive-by refactoring.
- **Tenant safety** — every organization-owned row carries `organization_id`; cross-organization access must fail and be tested.
- **Process rules** — never implement on `main`; branch per task; conventional commits; draft PRs a human merges; never fabricate command output or test totals.
- **Security rules** — no secrets in the repo, no weakened tests, all AI-provider calls through the core AI Gateway only.

2 · The working loop

1. **Assess** — read module READMEs, migrations, neighboring code, and tests before proposing anything; record findings in `status.md` / `plan.md`.
2. **Design first** — schema-affecting work started as a reviewed Architecture Decision Record: **ADR-008 Historical enrolment** and **ADR-009 Teaching assignments** were authored, reviewed, and accepted before implementation.
3. **Implement on a branch** — one task, one branch, conventional commits.
4. **Verify proportionally** — targeted tests, the full suite, Laravel Pint, PHPStan level 5, and MySQL migration checks; exact totals reported in every PR. Final state: **274 tests / 1,537 assertions, CI green on every push**.
5. **Rehearse live** — the agent drove the running application over HTTP as each persona: a scripted **12-point end-to-end journey** (teacher creates a quiz → learner writes it → AI-assisted marking → release → study plan → parent report → guardian portal) had to pass against the real stack before push.
6. **Human merge** — a person reviewed and merged every PR. When GitHub's automatic branch updates silently dropped two view files, the discipline caught it: CI failed, the agent diagnosed the lost hunks, and restored them in a reviewed fix commit.

3 · What was delivered under this process

- **Historical enrolment** (ADR-008): date-ranged placement history with backfill and self-healing — report-card history survives mid-year class moves.
- **Teaching assignments** (ADR-009): date-ranged teacher-to-class/subject assignments, per-organization opt-in enforcement across marking, attendance and scheduling, with an administrator bypass.
- **Persona experience**: role-aware login routing and navigation (learner → quizzes, guardian → portal, teacher → assessments) and a design-system refresh of the demo journey.
- **AI with teacher oversight**: AI-suggested marks with rationale and misconceptions that never become official without teacher approval; a teacher-authored **report to parent**; a **deterministic study-plan fallback** so a released result always ships a plan, even offline.

- **Family visibility:** guardian portal, report-card publication notifications, and the learner's own published report-card page.
- **Platform hardening:** AI-gateway transport-error wrapping, UTF-8 scrubbing, audit records on every consequential action, cross-organization isolation tests.

4 · Why this matters

The point is not that an AI wrote code — it is that the AI worked inside an **auditable engineering system**: designs recorded as ADRs, tenancy and security rules enforced as written policy, every claim backed by runnable checks, and a human owning every merge. The same principle the product applies to AI marking — *AI suggests, the teacher decides* — governed how the product itself was built.